

SANGHOON MYUNG

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SUMMARY

Principal Engineer with 16+ years of experience bridging semiconductor device physics, TCAD modeling, and deep learning. Expert in physics-informed ML, compact modeling, hybrid TCAD–AI simulation, and semiconductor process/device optimization. Published in top-tier venues including ICML, NeurIPS, IEDM, and VLSI, as well as leading journals in device modeling, with multiple US patents.

Core competencies:

Physics-informed machine learning, TCAD simulation, predictive modeling, TCAD–AI hybrid modeling, technical leadership, semiconductor device and process design

PROFESSIONAL EXPERIENCE

Principal Engineer —Semiconductor R&D Center, Samsung Electronics

2010 - Present

Deep Learning Researcher

2018 - Present

- Lead research on bridging deep learning and semiconductor physics for device and compact model development.
- Developed automated pipelines for TCAD data generation and AI model training.
- Proposed a novel calibration method aligning AI models with real measurements.
- Designed and deployed process digital twin models integrating TCAD, measurement data, and deep learning to enable virtual process optimization and predictive what-if analysis.
- Led a team of 4-5 engineers in model development and deployment.
- **Key Achievement:** Published in ICML, VLSI, IEDM, NeurIPS, and contributed multiple US patents.

TCAD Device Engineer

2010 - 2017

- Led TCAD design for CMOS Image Sensor (BDTI, 1.12 μm pixel), noise source modeling, and 3-layer stacked sensor design.
- Designed power devices (Logic 90 nm, BCD 130 nm) and created PDKs for PMIC development.
- Optimized implantation and junction engineering for SRAM VT control and yield prediction.
- Contributed to the successful development of Samsung's first 14 nm FinFET for mobile processors, including well isolation and process optimization.
- Led a team of 4-5 engineers in model development and deployment.
- **Key Achievement:** Awarded task incentives for 14-nm FinFET device development.

EDUCATION

BS in Electronic and Electrical Engineering (GPA: 3.9/4.5) — Kookmin University March 2003 - February 2010

AWARDS

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|---|----------------|
| - Grand Prize — DS CTO Symposium | November 2025 |
| - Excellence Prize — DS AI BP Festival | September 2025 |
| - Innovation Awards — Innovation Center | May 2022 |
| - Best Paper Awards — DIT Center | December 2021 |
| - Task Incentives for etching V6 channel hole — Semiconductor R&D Center | September 2019 |
| - Grand prize — Samsung Data Science Festival | October 2019 |
| - Silver Prize — Semiconductor R&D e-journal | September 2019 |
| - Task Incentives for 14nm Logic FinFET Devices — Semiconductor R&D | September 2012 |

TALK

- **Long Talk in Samsung Digital Twin Forum (50mins)**
— "Science Meets AI" *July 2025*
- **Long Talk in Symposium on VLSI Technologies and Circuits (25mins)**
— "A New Industry Standard Compact Model Integrating TCAD into SPICE" *June 2024*
- **Invited Talk in International Conference on Solid State Devices and Materials (30mins)**
— "Opportunity of Deep Learning Applicable to TCAD" *October 2022*
- **Short Talk in International Conference on Machine Learning (5mins)**
— "PAC-Net: A Model Pruning Approach to Inductive Transfer Learning" *July 2022*
- **Long Talk in International Electron Device Meeting (25mins)**
— "Restructuring TCAD System: Teaching Traditional TCAD New Tricks" *December 2021*
- **Short Talk in Conference on Neural Information Processing Systems (7mins)**
— "A Novel Method for Semiconductor Etching Process with Inductive Biases" *December 2020*
- **Long Talk in International Conference on Simulation of Semiconductor Processes and Devices (20mins)**
— "Real-Time TCAD: a new paradigm in the artificial intelligence era" *October 2020*

TECHNICAL SKILLS

Programming:	Python, L ^A T _E X, Verilog-A
ML Frameworks:	Tensorflow, Pytorch
Tools:	Synopsys TCAD, HSPICE
Database:	MySQL, SQLite.
Software & Tools:	Synopsys TCAD, HSPICE

ADDITIONAL ACTIVITIES

- **Organizing Workshop at Symposium on VLSI Technology**
— Planning and executing a digital twin workshop entitled "Innovating Semiconductor Manufacturing: Science Meets AI", focusing on applications and benefits. *June 2025*
- **Reviewer in Semiconductor Devices**
— Transaction on Electron Devices (TED), Electron Device Letters (EDL), Transactions on Semiconductor Manufacturing (IEEE TSM), Transaction on Computers, and Solid State Electronics (SSE) *October 2021 - 2025*
- **Reviewer in Artificial Intelligence**
— International Conference on Machine Learning (ICML), Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), and Artificial Intelligence and STATistics (AISTATS) *October 2020 - present*
- **Reviewer in Samsung Group**
— Samsung Best Paper Awards (Base Technology, Artificial Intelligence) *June 2021 - 2024*
- **Lecturer**— TCAD simulation, Logic Process, Deep Learning, Transfer Learning *March 2011 - present*
- **Military Service**— Signal corpsman *October 2005 - October 2007*

PUBLICATION

* indicates corresponding author

Conference

1. H. Jang, **S. Myung**, et al "Design-aware full-chip warpage modeling for stco: Bridging reliability and design for a new era of advanced systems", In *Symposium on VLSI Technologies and Circuits*, 2025
2. Y. Choi, **S. Myung***, et al "Real Time TCAD Calibration via Transfer Learning", In *International Conference on Simulation of Semiconductor Processes and Devices*, 2024
3. K. Kim, **S. Myung***, et al "Neural Drift-Diffusion Model Based on Operator Learning in Fourier Space", In *International Conference on Simulation of Semiconductor Processes and Devices*, 2024
4. **S. Myung***, et al "A New Industry Standard Compact Model Integrating TCAD into SPICE", In *Symposium on VLSI Technologies and Circuits*, 2024
5. (Invited) C. Jeong, **S. Myung**, et al "Deep Learning for Semiconductor materials and Devices Design", In *Electron Devices Technology and Manufacturing*, 2023
6. (Invited) **S. Myung***, et al "Opportunity of Deep Learning Applicable to TCAD", In *International Conference on Solid State Devices and Materials*, 2022

7. **S. Myung***, et al "PAC-Net: A Model Pruning Approach to Inductive Transfer Learning", In *International Conference on Machine Learning*, 2022
8. **S. Myung***, et al "Restructuring TCAD System: Teaching Traditional TCAD New Tricks", In *International Electron Device Meeting*, 2021
9. (Invited) C. Jeong, **S. Myung**, et al "Hybrid Modeling of TCAD and AI for Semiconductor Design", In *International Workshop on Computational Nanotechnology*, 2021
10. **S. Myung***, et al "A Novel Approach for Semiconductor Etching Process with Inductive Biases", In *Neural Information Processing Systems Workshop*, 2020
11. **S. Myung***, et al "Real-Time TCAD: a new paradigm for TCAD in the artificial intelligence era", In *International Conference on Simulation of Semiconductor Processes and Devices*, 2020
12. Y. kim, **S. Myung**, et al "Physics-augmented Neural Compact Model for Emerging Device Technologies", In *International Conference on Simulation of Semiconductor Processes and Devices*, 2020

Journal Article

1. **S. Myung***, et al "Simulation-guided AI-driven digital twin for plasma etching in semiconductor fabrication", In *Device by Cell Press*, 2026
2. W. Jang, **S. Myung***, et al "TCAD Device Simulation with Graph Neural Network", In *Electron Device Letters*, 2023
3. (Invited) **S. Myung***, et al "Comprehensive Studies on Deep Learning Applicable to TCAD" In *Japanese Journal of Applied Physics*, 2023
4. (Invited) C. Jeong, **S. Myung**, et al "Bridging AI and TCAD : its application to semiconductor design", In *Transaction on Electron Devices*, 2021

Patents

1. S. Myung, et al (2025) "Apparatus and method for performing process simulation of semiconductor devices", US Patent App. 19/024,261
2. K. Kim, S. Myung, et al (2025) "Simulation method and system based on machine learning model with reduced training requirements", US Patent App. 19/024,509
3. K. Kim, S. Myung, et al (2025) "Auto calibration method and system", US Patent App. 18/948,987
4. S. Myung, et al (2025) "METHOD AND COMPUTING DEVICE FOR GENERATING NEURAL NETWORK MODEL AND PERFORMING CIRCUIT SIMULATION BY USING THE NEURAL NETWORK MODEL", US Patent App. 18/768,183
5. W. Jang, **S. Myung**, et al (2024) "Semiconductor device simulation system and method", US Patent App. 18/099,083
6. S. Myung, et al (2024) "Simulation system for semiconductor process and simulation method thereof", US Patent 11,886,783
7. W. Jang, S. Myung, et al (2023) "Non-transitory computer-readable medium storing program code generating wafer map based on generative adversarial networks and computing device including the same" US Patent 11,775,840
8. J. Kim, **S. Myung**, et al (2023) "Computing device and method generating optimal input data", US Patent App. 17/994,089
9. **S. Myung**, et al (2023) "Method and system for three-dimensional modeling", US Patent App. 17/972,809
10. S. Shin, **S. Myung**, et al (2023) "Method for generating model using virtual target and system for the same", US Patent App. 17/861,582
11. **S. Myung**, et al (2023) "Simulation system for semiconductor process and simulation method thereof", US Patent 11,574,095
12. **S. Myung**, et al (2023) "METHOD AND APPARATUS FOR GENERATING PROCESS SIMULATION MODELS", US Patent App. 17/852,024
13. J. Kim, **S. Myung**, et al (2022) "Simulation method for semiconductor fabrication process and method for manufacturing semiconductor device", US Patent App. 17/230,275
14. I. Huh, **S. Myung**, et al (2021) "Computing device, operating method of computing device, and storage medium", US Patent App. 17/907,708
15. C. Jeong, **S. Myung**, et al (2021) "Method and system for hybrid model including machine learning model and rule-based model", US Patent App. 16/910,908
16. C. Jeong, **S. Myung**, et al (2020) "Semiconductor fault analysis device and fault analysis method thereof", US Patent 10,650,910